

Quiz 4

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (5 points) Given

$$\mathbf{A} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 4 & -4 \\ -1 & 3 & -1 \end{bmatrix}.$$

Find $\mathbf{A}^{-1}$ using the Gaussian-Jordan Elimination.

Problem 2. (5 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 0 & -3 \\ 0 & -1 & 1 \\ -4 & 5 & 3 \end{bmatrix}$. Find the **LU** and **LDU** factorizations of $\mathbf{A}$.